Meter Gomoku Link · 使用说明

五子棋连线训练器 版本 4.37 作者：Meter

一、软件简介

Meter Gomoku Link（五子棋连线训练器）是一款辅助五子棋对弈的桌面工具。它通过屏幕识别读取你正在进行的棋局，调用原生 Rapfi 引擎思考，并可在你授权后自动落子。识别在连线窗完成，引擎在软件本体后台常驻，局面、执子与思考时间自动同步到主窗大棋盘。软件与引擎之间采用进程内 C++ 直连，不经过任何网络端口（零 HTTP、零 WebSocket），全程本地运行、无需联网。

二、快速上手

（1）启动软件后，点击顶部「连线」键，打开连线窗口。

（2）点击「框选棋盘」，用鼠标在屏幕上框出目标棋盘所在的区域。

（3）点击窗栏的「识别」键；识别成功后选窗变为蓝绿色，四角出现 L 形角框与定位点，连线窗小棋盘与主窗大棋盘同时摆出识别到的局面。

（4）在「AI 执黑」或「AI 执白」中选择让 AI 持有的一方，二者只能选其一；选择会同步刷新主窗对局设置中的执子。

（5）需要自动对弈时点击「自动落子」；若只想获得提示，可不打开自动落子，按推荐点自行落子。

三、框选棋盘与识别

（1）框选：点击「框选棋盘」后，按住鼠标左键拖出矩形，把目标棋盘完整包含在内，可适当多框一些边缘。

（2）中空选窗：松开鼠标后出现一个物理中空的矩形窗口，你可以透过它直接点击到下方棋盘。

（3）持续识别：识别采用帧间像素差分看门算法（与单机版五子棋助手同一套方案）——棋盘画面静止时零消耗，棋子变化时才重新识别；稳定后才采纳，绝不让单帧抖动影响局面。

（4）成功标志：选窗变为蓝绿色；棋盘四角出现角框与定位点；识别范围自动向外扩约 0.6 个格子，保证边缘棋子读得完整。

（5）微调：可拖动中空选窗边缘微调范围；微调后软件只在该范围内识别，不再自动到屏幕别处寻找棋盘。

（6）关闭：关闭中空选窗后，识别暂停；再次点击「框选棋盘」可重新开始。

四、AI 执黑、AI 执白与思考时间

（1）「AI 执黑」与「AI 执白」为变色按键，选中后高亮，二者互斥：选黑则白取消，选白则黑取消。

（2）执子与思考时间都会无条件同步到主窗对局设置：连线窗选了什么，主窗就是什么，绝不会出现双方都被评估的情况。

（3）「思考时间」用于设置引擎每一步的计算时长，默认 2000 毫秒；软件更强调在固定时间内把算力用到极致，而非单纯堆时间。

（4）当思考时间设为 1 秒及以下时，软件会在计算期间临时向 CPU 申请更高频率，算完后恢复，以改善短时间的棋力。

五、自动落子

（1）打开「自动落子」后，软件识别到对方落子即自动思考，并采用三级模拟点击管线（PostMessage → SendInput → 真实点击）在目标棋盘上替你落子，不影响你的物理鼠标。

（2）若你是先手（AI 执黑、空棋盘），打开自动落子后会自动下出第一步（天元）。

（3）落子是否真的成功，由扫描回帧裁决：没落上会自动升级点击管线重试，连续失败则停机并提示。

（4）每完成一局（出现五子连珠），自动落子会自动关闭，下一局需重新点击。

（5）关闭自动落子时，软件只显示推荐落点，不替你操作鼠标。

六、交换平衡点与交换评估

（1）交换平衡点：用于交换规则开局，寻找让黑白双方都近乎均势（评估接近 0 分）的落点。空棋盘时给出三个点（两黑一白）；棋盘上已有棋子时给出两个点（一黑一白）；棋盘上超过五个棋子时不再调用。

（2）点击交换平衡点时，会自动取消已选择的 AI 执黑 / 执白（避免相互干扰）；评估完成后评估点留在识别框中，平衡评估引擎随即关闭、不再占用算力。

（3）当你在棋盘上下完这几个子后，无论落在哪里，评估标记都会自动消失。

（4）顺序逻辑：若先点击交换平衡点、后打开自动落子，软件会在评估完成后按顺序自动把这几个平衡点落完 —— 与单机版五子棋助手完全一致。

（5）交换评估：与交换平衡点相互独立的另一项功能，用于评估当前交换局面的优劣，供你判断是否交换。

七、连线窗与主窗的分工

（1）连线窗负责识别与 AI 的全部工作（评估、自动落子）；主窗大棋盘只同步显示识别到的局面与执子设置，本身不参与评估、不落子 —— 两侧不会互相抢着计算。

（2）识别到的每一手棋都同步摆上主窗大棋盘；连线窗小棋盘与助手的局面小窗同款语义：识别到什么就显示什么，没识别到就保持上一手好盘面，不会无故清空。

（3）关闭连线窗后，主窗自动恢复你自己原来的模式与设置，对弈 / 计算等功能照常使用。

八、标记显示

（1）推荐落点以天蓝色圆环直接画在识别框内的对应交点上，置于系统最上层、且不会被截图识别，避免被误读为棋子。

（2）交换平衡点评估完成后，黑方为紫色圆环、白方为蓝色圆环，并带有 1、2、3 的序号数字，同样画在识别框内。

九、规则与外观

（1）支持无禁手（自由局）、长连不赢、有禁手（连珠）等规则，识别与算杀均跟随所选规则。

（2）界面支持中文 / 英文一键切换，并可适配深色与浅色主题；软件文字 logo 恒为英文「Meter Gomoku Link」。

十、常见问题

（1）识别不到棋盘：请重新「框选棋盘」，确保范围完整包含棋盘四角与边框；棋子较满时，软件会结合棋子与边框共同判断。

（2）一直提示正在识别：可关闭中空选窗后重新框选；首次打开时软件会进行识别预热，稍候即可。

（3）关闭后是否残留进程：软件退出时会一并关闭所有引擎与相关进程，不在后台残留。

（4）是否需要联网：不需要，软件与引擎全部在本地运行。

作者：Meter 本软件界面与配套代码以 MIT 许可发布，棋力内核 Rapfi 以 GPL-3.0 发布，文档排版使用思源宋体 / 思源黑体（SIL OFL 1.1 开源授权）。开源详情见随附《开源说明》。

Meter Gomoku Link · User Guide

Five-in-a-Row Link Trainer Version 4.37 Author: Meter

1. Introduction

Meter Gomoku Link is a desktop assistant for Gomoku (Five in a Row). It reads the game on your screen, thinks with the native Rapfi engine, and can play moves for you when you allow it. Recognition happens in the Link window while the engine runs in the background of the main application; the position, the AI side and the thinking time are mirrored onto the main board. The UI and the engine talk to each other in-process through C++ only, with no network ports at all (zero HTTP, zero WebSocket). Everything runs locally - no Internet connection is required.

2. Getting Started

(1) Launch the app and click "Link" on the top bar to open the Link window.

(2) Click "Select board" and drag a rectangle over the target board on screen.

(3) Press "Recognize" on the title bar. On success the box turns teal, L-shaped corner brackets and anchor points appear, and both the mini board and the main board show the recognised position.

(4) Choose either "AI plays Black" or "AI plays White" (only one can be active). The choice is mirrored into Game Setup in the main window.

(5) Click "Auto drop" for automatic play, or leave it off and follow the suggested moves yourself.

3. Selecting and Recognizing the Board

(1) Select: click "Select board", hold the left mouse button and drag a rectangle that fully contains the board, leaving a little margin.

(2) Hollow window: a physically see-through rectangle appears, so you can click the board underneath through it.

(3) Continuous recognition: the app uses an inter-frame pixel-diff watch (the same scheme as the standalone assistant) - nothing is computed while the board is still, and a re-read happens only when the stones change. A position is adopted only once it is stable, so a single noisy frame never disturbs your game.

(4) On success the box turns teal, corner brackets and anchors appear, and the recognised area expands outward by about 0.6 of a cell so edge stones are read completely.

(5) Adjust: drag the edges of the hollow window to fine-tune the area; afterwards the app only recognises within it and no longer searches the rest of the screen.

(6) Close: closing the hollow window pauses recognition; click "Select board" again to restart.

4. AI Side and Thinking Time

(1) "AI plays Black" and "AI plays White" are colour-toggle buttons that are mutually exclusive: choosing one cancels the other.

(2) Both the side and the thinking time are mirrored unconditionally into Game Setup in the main window, so the two can never disagree and the engine never evaluates both sides at once.

(3) "Thinking time" sets how long the engine computes each move (default 2000 ms). The app focuses on using the fixed time to the fullest rather than simply adding time.

(4) At 1 second or below, the app briefly requests a higher CPU frequency during the calculation and restores it afterwards to strengthen short-time play.

5. Auto Drop

(1) With "Auto drop" on, once the opponent moves the app thinks and plays for you through a three-stage simulated click pipeline (PostMessage, then SendInput, then a real click) that never disturbs your physical mouse.

(2) If you play first (AI Black, empty board), turning it on plays the opening move at the centre (Tengen).

(3) Whether a drop really landed is judged from the next recognised frame: if it did not land, the click pipeline is escalated and retried; after repeated failures it stops and tells you.

(4) After each finished game (five in a row) Auto drop switches off; turn it on again for the next game.

(5) With Auto drop off, the app only shows the suggested move and never moves your mouse.

6. Swap Balance and Swap Evaluation

(1) Swap Balance finds opening moves that keep both sides almost equal (an evaluation near zero) under swap rules. On an empty board it gives three points (two Black, one White); with stones present it gives two (one Black, one White); it is not used once more than five stones are on the board.

(2) Pressing Swap Balance automatically cancels any selected AI side first (they would interfere). When the evaluation finishes the markers stay in the recognition frame and the balance engine is released immediately, so no computing power is wasted.

(3) Once you have played those stones the markers disappear automatically, wherever they were played.

(4) Order of operations: if you press Swap Balance first and Auto drop is on when the evaluation finishes, the points are played automatically in order - exactly like the standalone assistant.

(5) Swap Evaluation is a separate, independent feature that scores the current swap position to help you decide whether to swap.

7. Link Window and Main Window

(1) The Link window owns all recognition and all AI work (evaluation, auto drop). The main board only mirrors the recognised position and the AI-side settings; it never evaluates and never drops - the two can never fight over the same search.

(2) Every recognised move is mirrored onto the main board. The mini board follows the same rule as the assistant position window: it shows what the current frame recognised, and keeps the last good position when nothing is recognised - it is never cleared for no reason.

(3) Closing the Link window restores your own mode and settings in the main window; play, computation and everything else keep working as before.

8. On-board Marks

(1) The suggested move is drawn as a sky-blue ring directly on the matching intersection inside the recognition frame, on the topmost system layer and excluded from screen capture, so it is never misread as a stone.

(2) After a Swap Balance evaluation, Black points are purple rings and White points are blue rings, numbered 1, 2, 3 - drawn inside the recognition frame as well.

9. Rules and Appearance

(1) Free-style (no forbidden), overline does not win, and Renju (forbidden moves) rules are supported; recognition and mates follow the chosen rule.

(2) The UI switches between Chinese and English in one click and supports dark and light themes. The product wordmark is always English: "Meter Gomoku Link".

10. Frequently Asked Questions

(1) Board not recognised: re-run "Select board" and make sure the four corners and the border are fully enclosed; on a crowded board the app combines the stones and the border to decide.

(2) Stuck on "recognizing": close the hollow window and re-select; the app warms up recognition on first open, so please wait a moment.

(3) Leftover processes after closing: when the app exits it shuts down every engine and related process, leaving nothing behind.

(4) Is Internet required: no; the app and the engines run entirely on your computer.

Author: Meter. The interface and companion code are released under the MIT licence; the playing core Rapfi is released under GPL-3.0; the manuals are typeset in Source Han Serif / Source Han Sans (SIL OFL 1.1). See the accompanying Open-Source Notice for details.